

Project ID :

25-26J-294

1. Topic (12 words max)

Smart Helmet for Cyclists and Motor Cyclists

2. Research group the project belongs to

SST - Software Systems & Technologies

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	25-26J-294
Year	2025

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

Cyclists and motorcyclists are frequently exposed to road hazards, unexpected health conditions, and psychological stress, which traditional helmets are not equipped to manage. While conventional helmets provide passive protection, this research proposes a smart helmet system that actively monitors physiological and cognitive states to prevent emergencies and promote safe, stress-free riding.

The system integrates biometric and motion sensors—including heart rate, temperature, EEG (brainwave), and IMU (motion)—into a compact, wearable helmet. Using IoT and machine learning technologies, the helmet continuously tracks the rider's heart rhythm, stress levels, and motion anomalies (e.g., falls, sharp turns). Real-time alerts are delivered through voice commands or mobile app notifications, enabling riders to respond instantly. For example, if signs of a potential heart attack or severe stress are detected, the system can advise rest, initiate breathing exercises, or allow hands-free emergency communication (e.g., "Send Help").

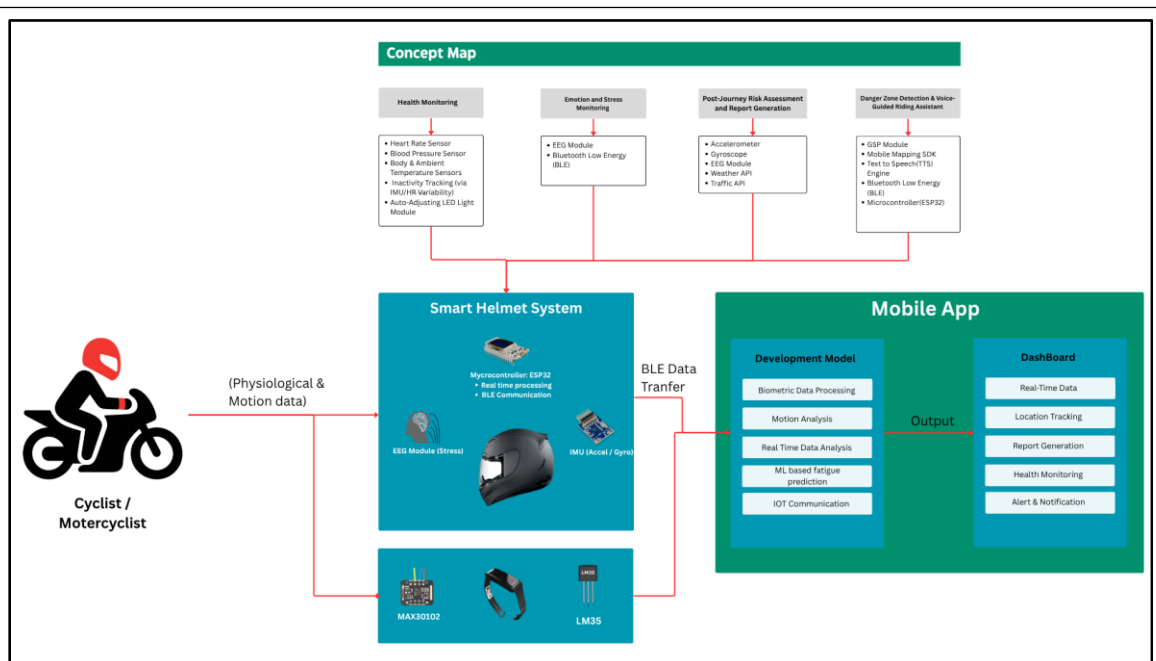
Beyond live monitoring, the helmet generates personalized post-ride analytics. It highlights segments involving risky behaviors such as speeding under stress, riding through harsh weather, or signs of cognitive fatigue. This approach leverages predictive models that combine real-time and historical sensor data to forecast fatigue and stress-related health risks—helping riders avoid burnout, mental overload, or cardiovascular strain.

The concept is grounded in findings from WHO reports on road safety [1] and supported by research into wearable health monitoring and machine learning–driven risk prediction [2][3][4][5]. Notably, prior studies have demonstrated the potential of IoT-integrated wearable devices for heart attack prediction and mental state detection [9][10][11], underscoring the relevance of this approach. This smart helmet redefines personal safety by shifting from reactive protection to proactive, intelligent intervention, empowering riders with insight, awareness, and autonomy.

References

1. World Health Organization. Global status report on road safety 2018. <https://www.who.int>
2. P. Casson et al., "Wearable EEG: What is it, why is it needed, and what does it entail?," IEEE Signal Processing Magazine, vol. 36, no. 1, pp. 38–47, 2019.
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6. Pathak, P., & Chaurasia, V. (2021). *Prediction and detection of heart attack using AI and ML technology*. International Journal of Engineering Research & Technology (IJERT), 10(05), 1–5. <https://www.ijert.org/prediction-and-detection-of-heart-attack-using-ai-and-ml-technology>
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11. Perera, K. (2021). *Smart helmet accident avoidance system* [Master's thesis]. CORE. <https://core.ac.uk/works/129952402>
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6. Brief description of the nature of the solution including a conceptual diagram



This project presents an intelligent smart helmet system designed to enhance rider safety and well-being by integrating real-time health monitoring, emotional state tracking, danger zone alerts, and post-ride risk analysis. Unlike conventional helmets, this solution actively monitors the rider's physical and mental state using low-cost, wearable biosensors and provides proactive feedback to prevent accidents and long-term health risks.

The system includes sensors such as heart rate, body temperature, EEG (for stress detection), and IMU components (accelerometer, gyroscope, magnetometer) to monitor health, emotional strain, and riding behavior. An ESP32 microcontroller processes sensor data on the edge and transmits it to a mobile application via Bluetooth Low Energy (BLE). Voice recognition enables hands-free emergency actions like "Send Help" or "Call Contact" in critical health scenarios.

Before rides, the helmet identifies danger zones based on historical route data and visualizes them in the app using spatial indexing (R-tree/KD-tree). During rides, real-time GPS tracking and text-to-speech voice alerts notify the rider of nearby risk zones in their selected language (English or Sinhala). Emotionally adaptive feedback and relaxation tips are provided based on EEG analysis.

After each ride, the system compiles journey data with external weather and traffic API inputs to generate a personalized risk report highlighting unsafe segments, stress peaks, and behavioral patterns. Weekly summaries help identify chronic health or stress concerns early. This intelligent system transforms the helmet into an active health companion, bridging the gap between personal mobility and proactive healthcare monitoring. It is grounded in global road safety statistics (WHO) and supported by research in IoT-based health monitoring (IEEE).

7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

The development of this smart helmet system requires multidisciplinary expertise across embedded systems, biomedical sensing, signal processing, wireless communication, machine learning, mobile development, and geospatial analysis. Core hardware knowledge includes integration of physiological sensors (heart rate, temperature, EEG), motion sensors (accelerometer, gyroscope, magnetometer), and communication modules (ESP32 with BLE/Wi-Fi). Proficiency in embedded programming using Arduino IDE and ESP-IDF is essential for real-time data acquisition and edge analytics.

Expertise in EEG signal processing and brain-computer interface (BCI) design is required to extract emotional and cognitive stress patterns for real-time feedback. Additionally, skills in time-series analysis and anomaly detection are vital to detect cardiac risk, sudden movements, or fall events. The implementation of predictive algorithms for post-ride risk profiling requires knowledge of multimodal data fusion, context-aware analysis, and behavioral modeling.

Mobile app development skills are needed to build an intuitive interface that supports real-time monitoring, voice alerts, multilingual interaction, and visualized health/risk summaries. Integration with offline voice recognition engines (e.g., PicoVoice) and cloud backends (e.g., Firebase) is necessary for hands-free commands, data storage, and weekly report generation.

The system also relies on access to quality datasets—such as labeled EEG stress data, heart rate variability (HRV) logs, and IMU-based riding behavior profiles—for model training and validation. External APIs like OpenWeatherMap and Google Traffic provide contextual data for enhancing post-ride reports.

Toolchains include TensorFlow Lite (for on-device ML), Firebase (cloud sync and user data), Arduino IDE (firmware development), Mapbox SDK (route visualization), and R-tree/KD-tree libraries (for spatial indexing).

This interdisciplinary blend of domains transforms a basic helmet into an AI-powered, health-aware safety assistant that supports preventive, responsive, and personalized rider care.

8. Objectives and Novelty

Main Objective The main objective of the Smart Helmet system is to enhance rider safety and well-being by continuously monitoring vital signs and stress levels, providing real-time alerts, hands-free emergency communication, and personalized feedback to prevent accidents and promote healthy, stress-free journeys			
Member Name with Registration No	Sub Objective	Tasks	Novelty
Vithanage K.W.Y.L.N (IT22595980)	Health Monitoring System	- Integrate heart rate and temperature sensors into the helmet. - Develop firmware for ESP32 to read sensor data, process it in real time, and transmit it to the mobile app. - Implement a voice recognition system for hands-free emergency commands (e.g., “Call Contact”, “Send Help”). - Design and code mobile app features for live health tracking, alerts, and personalized journey-based advice. - Create algorithms to detect cardiac risk, analyze health data, emotional and trigger warnings or suggestions based on detected trends.	Predict heart attack risk using multi-sensor health and stress data. Enable hands-free emergency response via voice commands. Offer context-aware health coaching during rides. Provides weekly health trend reports for early risk detection. Uses edge analytics on ESP32 for fast, real-time alerts.

		- Enable the system to generate voice-guided, context-aware health advice during the ride. - Provide personalized coaching and positive reinforcement to promote safe riding behavior and long-term cardiovascular health awareness. - Visualizing the medical report as a graph. Using major critical points.	
Samarakoon S.S.A.D.S.B (IT22207036)	Emotion and Stress Monitoring with Real-Time Feedback	- Integrate an EEG module to continuously capture brainwave data during rides. - Develop BLE (Bluetooth Low Energy) communication to send EEG data wirelessly to the mobile app. - Build and train a stress classification model to analyze EEG patterns for stress detection. - Implement real-time mobile notifications and recommendations (e.g., suggest breaks, breathing exercises, or playing calm music) based on detected stress levels. - Display live emotional state and stress alerts in the app UI.	The system offers real-time monitoring of the rider's emotional and mental stress levels using EEG data. A trained stress classification model detects individual stress patterns and triggers instant alerts or personalized suggestions such as breaks or relaxation cues. EEG data is transmitted seamlessly via BLE, and the rider's current emotional state is displayed in the app.
Samarasinghe K.P.C (IT22608086)	Post-Journey Risk Assessment and Report Generation	- Collect and synchronize journey data from IMU sensors (accelerometer, gyroscope, magnetometer) and EEG during the ride.	The system provides personalized, context-aware risk reports by combining motion and EEG data with real-time weather and traffic

		- Analyze data to detect risky behaviors: high-speed events, sudden movements, sharp turns, and stress peaks. - Collect external data using Weather and Traffic APIs based on the rider's GPS route. - generate structured post-ride, user-friendly report in the mobile app summarizing high-risk zones, stress peaks, and potential danger moments. - Highlight critical events such as high-speed stress in poor weather or sharp turns in traffic zones. - Enable historical data logging for ongoing safety improvement.	information. It identifies risky behaviors influenced by environmental and emotional factors, offering clear insights to help riders understand and improve their safety habits.
Primasha W.G.R (IT22216878)	Danger Zone Detection & Voice-Guided Riding Assistant	- Use spatial indexing (R-tree/KD-tree) to quickly match upcoming route segments with risk zones. - Visualize danger zones on the mobile app map using heatmaps and warning icons via Map box or Google Maps API. - Enable user route input and compare it against historical data for proactive risk detection. - Implement real-time GPS tracking to determine rider's current position during the ride. - Trigger personalized voice alerts using TTS (Google TTS/Amazon Polly) as the rider approaches a danger zone.	The system uses multi-sensor data (GPS, IMU, EEG) to create personalized danger zone maps, enhancing rider safety through real-time, multilingual voice alerts and distance-aware risk announcements. It offers pre-ride risk forecasting with safer route suggestions and employs spatial indexing (R-tree/KD-tree) for fast, efficient detection of hazards along the ride path.

		- Add multilingual support in the app to select English or Sinhala for voice alerts. - Generate pre-ride safety tips based on route analysis and suggest safer alternate paths. - Deliver voice guidance detailing distance to the next danger zone and recommended actions (e.g., slow down, take alternate route). - Sync helmet processor (ESP32/Pico W) with mobile app via BLE for real-time updates.	
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9. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
Vithanage K.W.Y.L.N (IT22595980)	Developed a health monitoring system using heart rate, temperature, and EEG sensors integrated via ESP32. Implemented real-time cardiac risk detection, context-aware voice guidance, and hands-free emergency commands, demonstrating skills in embedded systems, IoT, and personalized health analytics.
Samarakoon S.S.A.D.S.B (IT22207036)	Implemented real-time emotion and stress monitoring using EEG sensors and BLE. Built a stress classification model and integrated live feedback features into the mobile app, applying expertise in brain-computer interface (BCI), wireless communication, and real-time health response systems.
Samarasinghe K.P.C (IT22608086)	Designed a post-journey risk assessment module that combines IMU and EEG data with weather and traffic APIs. Developed a system for generating detailed, context-aware safety reports, showcasing skills in sensor fusion, data analytics, and app-based report generation. Generate structured post-ride, user friendly report (Daily, Weekly, Monthly) in the mobile app summarizing high-risk zones, stress peaks, and potential danger moments. - Highlight critical events such as highspeed stress in poor weather or sharp turns in traffic zones.
Primasha W.G.R (IT22216878)	Before each ride, Check if route has been previously traveled. If the route to travel before, analyzes historical GPS, IMU, and EEG data to identify danger zones and displays them on the mobile app before starting the ride. If the route to travel is not familiar, it enhances route safety by integrating public accident databases, road hazard maps, weather condition APIs, and crowdsourced traffic data, enabling it to predict and highlight danger zones. Real-time GPS tracking triggers personalized voice alerts, while pre-ride tips and safer route suggestions promote informed decision-making and encourage safer riding behavior.

10. Supervisor's details

	Title	First Name	Last Name	Signature
Supervisor				

Co-Supervisor				
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

This part is to be filled by the Topic Screening Staff members.

- a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes		No	
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- b) Does the proposed topic exhibit novelty?

Yes		No	
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- c) Do you believe they have the capability to successfully execute the proposed project?

Yes		No	
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- d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes		No	
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- e) Supervisor's Evaluation and Recommendation for the Research topic:

Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Staff Member's Name	Signature

***Important:**

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.